

← Riya Clg
11/20/24, 10:06 AM

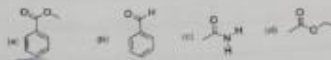
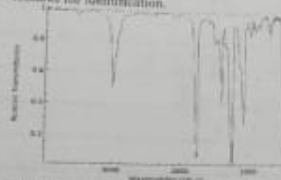


Indian Institute of Science Education and Research Mohali
CHM 201: INTRODUCTION TO SPECTROSCOPIC METHODS

Mid-Sem Examination-II, 24th October, 2024

Total marks: 20; Time: 10.45 PM – 11.45 PM

1. Consider a homonuclear diatomic molecule X_2 , the force constant of $X-X$ bond is ' k '. The atomic weight of X is ' A '. Show that the zero-point energy (ZPE) can be written as: (4 marks)
$$E_0 = \frac{h}{2\pi} \sqrt{\frac{k}{A}}$$
2. Rotational states don't show ZPE, whereas, vibrational states show ZPE. Why? Justify your answer. (3 marks)
3. The wavenumber of the incident radiation in a Raman spectrometer is $20,487 \text{ cm}^{-1}$. What is the wavenumber of the scattered Stokes radiation for the $J = 2 \leftarrow 0$ in $^{14}\text{N}_2$ ($k = 1.9956 \text{ cm}^{-1} \text{ Å}^{-1}$)? (3 marks)
4. Based on the IR spectral features, identify the correct compound among the given options. Also, assign the key features for identification. (1+2 marks)



5. (a) The table below lists the vibrational wavenumbers and infrared/Raman activities for cyanogen (C_2N_2). Assign the vibrational modes for each spectral line. (5+2 marks)

Frequency (cm^{-1})	Raman	IR	Assignments of vibrational modes
2122	Strong	-	
2149	-	strong, PR contour	
848	strong	-	
506	Very strong	-	
226	-	strong, PQR contour	

- (b) Draw the structure.

Universal constants: $N_A = 6.023 \times 10^{23} \text{ mol}^{-1}$; $k = 1.381 \times 10^{-23} \text{ J K}^{-1}$; $h = 6.626 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ m s}^{-1}$

